



REBOUND

BIOMIMETIC SONAR NAVIGATION SYSTEM

GLOBAL AI HACKATHON SERIES • MEMORYAGENT TRACK

NAVIGATE

EXPLORE THE PRESENTATION

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THE VISION

Inspiration, purpose, and what REBOUND does

02

HOW IT WORKS

Signal processing pipeline from chirp to guidance

03

ARCHITECTURE

Mobile-first design meets cloud intelligence

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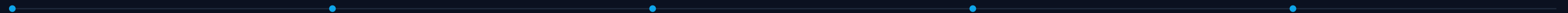
INTELLIGENCE

Adaptive memory that learns with every step

05

RESULTS

Empirically validated performance metrics



01

THE VISION

Reimagining navigation through nature's design



Inspired by Millions of Years of Evolution

REBOUND is a biomimetic sonar system for indoor navigation, designed for visually impaired users.

It draws inspiration from [bat CF-FM echolocation](#) — nature's most refined acoustic navigation system, developed over **50 million years** of evolution.

Built for the **Global AI Hackathon Series** with Qwen Cloud, competing in the [MemoryAgent](#) track.



BIOMIMETIC — Based on bat echolocation



ACOUSTIC — Ultrasonic chirp detection



ADAPTIVE — AI memory that learns

"Nature is the best engineer. We just had to listen — and build what we heard."

What REBOUND Does

REBOUND emits **ultrasonic chirps** from the device speaker, listens to echoes via the microphone, and classifies the acoustic environment in **real time**.

A Qwen-powered Memory Agent learns user behavior and personalizes navigation instructions.



EMIT

CF-FM chirp at 8 kHz through device speaker — the same technique bats use to navigate complete darkness



LISTEN

Capture echoes via microphone, deconvolve to obtain Room Impulse Response with Wiener filtering



GUIDE

Qwen Memory Agent generates personalized navigation instructions with haptic feedback via vibration

The result: a system that **doesn't just detect obstacles** — it understands context, remembers preferences, and guides with increasing precision over time.

02

HOW IT WORKS

From sound wave to intelligent guidance

The Signal Processing Pipeline

- 1 **EMIT** CF-FM chirp (8 kHz sweep) through speaker
- 2 **CAPTURE** echo and deconvolve to obtain RIR
- 3 **EXTRACT** mel spectrogram, MFCCs, RT60, centroid
- 4 **CLASSIFY** environment with PyTorch CNN (6 classes)
- 5 **DETECT** stairs via independent DSP detector
- 6 **REASON** with Qwen Memory Agent for navigation
- 7 **ADAPT** Bayesian priors across user sessions
- 8 **FEEDBACK** haptic response via device vibration

PIPELINE FEATURES

- ▶ **Real-time** processing on every observation
- ▶ **Wiener deconvolution** for clean RIR extraction
- ▶ **Mel spectrogram** 64x32 + **MFCCs** 13x32
- ▶ **RT60** reverberation time analysis
- ▶ **Spectral centroid** for tone characterization
- ▶ **Two-pass** periodic staircase detector
- ▶ **Episodic + semantic** memory updates

CNN Classes: open_space • nearby_wall • doorway • corner • corridor • stairs

Staircases: CNN class + independent DSP detector for mutual reinforcement

03

ARCHITECTURE

Mobile-first design meets cloud-scale intelligence

System Architecture

MOBILE (PWA)

- Web Audio API
- getUserMedia
- Speaker emit chirp
- DeviceOrientation
- Gyro pitch data
- navigator.vibrate()
- Haptic feedback



ALIBABA CLOUD ECS

POST /predict

- › Wiener deconvolution (RIR)
- › Feature extraction
- › CNN classifier (6 classes)
- › Stair detector (DSP)

POST /process

- › Qwen-Plus Memory Agent
- › Episodic/Semantic memory
- › User profile update
- › Bayesian prior adjustment



DashScope API (Qwen-Plus) — All agent calls are async via [httpx.AsyncClient](#) for concurrent users

GET /profile/{user_id} | POST /session | GET /health

GET /profile/{user_id} | POST /session | GET /health

Backend deployed on **Alibaba Cloud ECS** via Docker • FastAPI async server

Environment Classification

95.3%

VALIDATION ACCURACY

0.302_m

DISTANCE MAE

~27_{dB}

STAIRCASE SNR THRESHOLD

88/88

TESTS PASSING (100%)



OPEN SPACE

Unobstructed large area



NEARBY WALL

Close surface detected



DOORWAY

Passage between spaces



CORNER

Intersecting surfaces



CORRIDOR

Long narrow passage



STAIRS

CNN + DSP dual detection

Lightweight **PyTorch CNN** suitable for edge deployment
Training on **NVIDIA RTX 3090** • 20% stratified validation split

04

INTELLIGENCE

Adaptive memory that learns with every step



Qwen Memory Agent

COGNITIVE CAPABILITIES

- 1 **Reason** over sonar observations in natural language
- 2 **Decide** which memories to store, update, or forget (selective forgetting)
- 3 **Generate** personalized navigation instructions per user
- 4 **Adjust** Bayesian confidence priors based on user behavior

DUAL MEMORY SYSTEM



EPISODIC MEMORY

Session experiences stored as events. The agent remembers what happened, where, and when — building a personal navigation history for each user.



SEMANTIC MEMORY

Learned facts about environments and user preferences. Abstract knowledge that transfers across sessions — "doorways in this building echo differently."

A navigation experience that **truly adapts** to each individual —
learning, remembering, and evolving with every step.

05

RESULTS

Empirically validated across all metrics

SUCCESS INDICATORS

SUCCESS INDICATORS

SYSTEM PERFORMANCE
OPTIMAL

Performance Metrics

95.3%

CNN Validation Accuracy
6-class classification

0.302_m

Distance Estimation
Mean Absolute Error

88/88

Tests Passing
100% test coverage

~27_{dB}

Staircase Detector SNR
Threshold for detection

6

CNN Classes
Environment types



Training: NVIDIA RTX 3090
Edge-suitable lightweight model

Real-time classification with a **lightweight PyTorch model** for edge deployment,
while the full-memory agent runs in the cloud for scalable intelligence.

Security & Future Roadmap

SECURITY LAYERS

■ **user_id** validated with regex `[a-zA-Z0-9_\-]{1,64}`

Path traversal prevention

■ **class_name** validated against whitelist

Prompt injection prevention

■ LLM multipliers clamped to `[0.1, 10.0]`

Hallucination containment

■ Semantic memory capped at `512 chars`

Unbounded growth prevention

■ Per-user `asyncio.Lock`

Race condition prevention

FUTURE ROADMAP (V2)



ELASTIC WEIGHT CONSOLIDATION

Prevent catastrophic forgetting in the CNN when learning new environment types. Currently deferred to v2.



NOISE AUGMENTATION

Robust training data through synthetic noise injection. Known bug in current implementation — deferred to v2.

Note: No JWT/API key auth yet — deploy behind reverse proxy



REBOUND

Nature taught us how to see with sound.

AI gave us the memory to learn from every echo.

Designed & Developed by **Olga Vasilieva**

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